

Final examination on Logic Circuits Design (Group 2)

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1 Converting from decimal to negabinary

Question. The *negabinary* representation of a number is similar to the binary representation, except that the base is not 2 but -2 . Therefore, the shape of the negabinary representation with n bits $b_{n-1}, b_{n-2}, \dots, b_0$ is

$$b_{n-1}(-2)^{n-1} + b_{n-2}(-2)^{n-2} + \dots + b_1(-2)^1 + b_0$$

Devise a C function converting a decimal number, possibly negative, into its negabinary form:

```
int* from10toNeg2 (int dec);
```

where a negabinary number is represented as a pointer to an array of integers. For example, the implementation of the negabinary 1101_{-2} is a pointer to the leftmost cell of the array

4	1	1	0	1
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Hints.

1. In C, if $a \geq 0$ and $b < 0$, then $a/b \leq 0$ and $a \% b \geq 0$. Also, if $a \leq 0$ and $b < 0$, then $a/b \geq 0$ and $a \% b \leq 0$. For instance, $17 = (-8)(-2) + 1$ and $-17 = 8(-2) - 1$.
2. Compute the size `size` of the output with your algorithm first;
3. allocate the result array with length `size + 1`;
4. rerun the algorithm (a loop) but, this time, fix the bits and save them in the resulting array.